Practical aspects of training / validation

Predict Who Will Repay Their Loans

- A lender hires you to investigate who will repay their loans
  - You are given complete files on 100 applicants
  - 5 defaulted within 3 years

Image credit debt.org

A Surprising Finding

- All 5 people who defaulted wore blue shirts during interviews
- Your model may find this strong signal as well

Training Error and Generalization Error

- Training error (bias): model error on the training data
- Generalization error (variance): model error on new data
- Example: practice a future exam with past exams
  - Doing well on past exams (training error) doesn't guarantee a good score on the future exam (generalization error)
  - Student A gets 0 error on past exams by rote learning
  - Student B understands the reasons for given answers

Validation Dataset and Test Dataset

- Validation dataset: a dataset used to evaluate the model
  - Should not be mixed with the training data (#1 mistake)
  - Model selection purpose
- Test dataset: a dataset can be used once, e.g.
  - A future exam
  - The house sale price I bided
  - Dataset used in private leaderboard in Kaggle

Training/Validation/Test set

- Previous ML: 70/30
- Big data: 1,000,000 10,000 10,000
- Percentages: 98/1/1
- Percentages: 99.5/0.4/0.1

- K-fold CV is expensive for deep learning

Parameters vs Hyperparameters

Parameters : $W^{[1]}, b^{[1]}, W^{[2]}, b^{[2]}, W^{[3]}, b^{[3]} \ldots$

Hyperparameter:

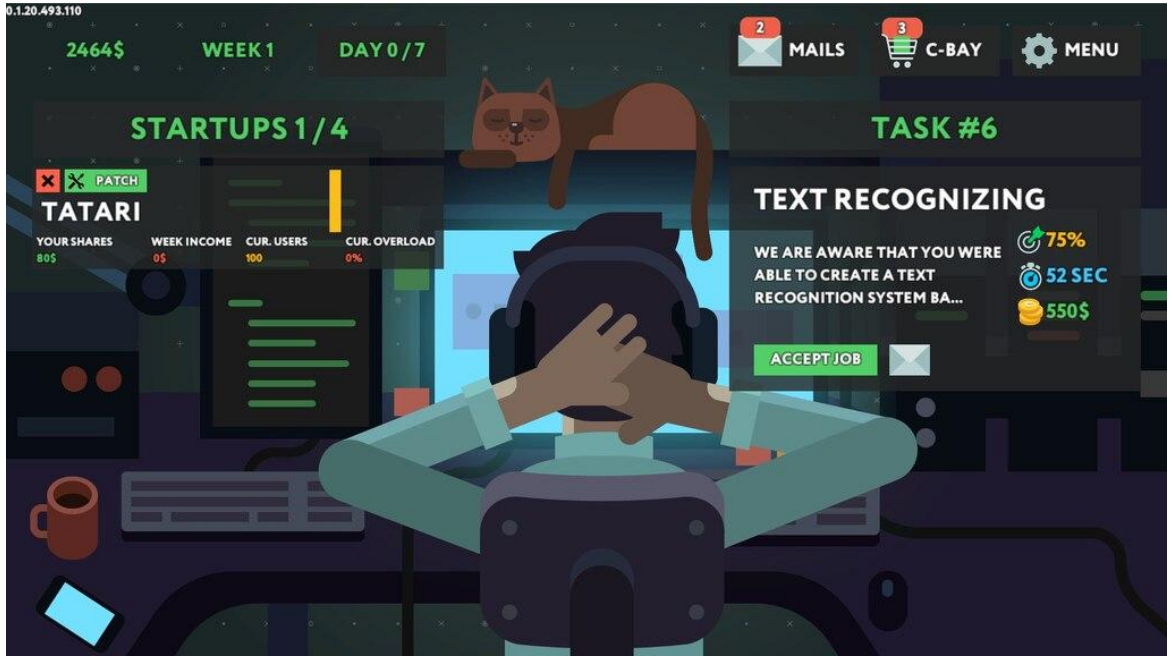
- Learning rate
- #epochs
- Batch size
- #hidden layers
- #hidden units
- Choice of activation function
- …

- Model selection

Summary

- Training dataset: train parameters with given hyper-parameters
- Validation dataset: choose hyper-parameters
- Test dataset: leave it alone

Applied DL is a very empirical process

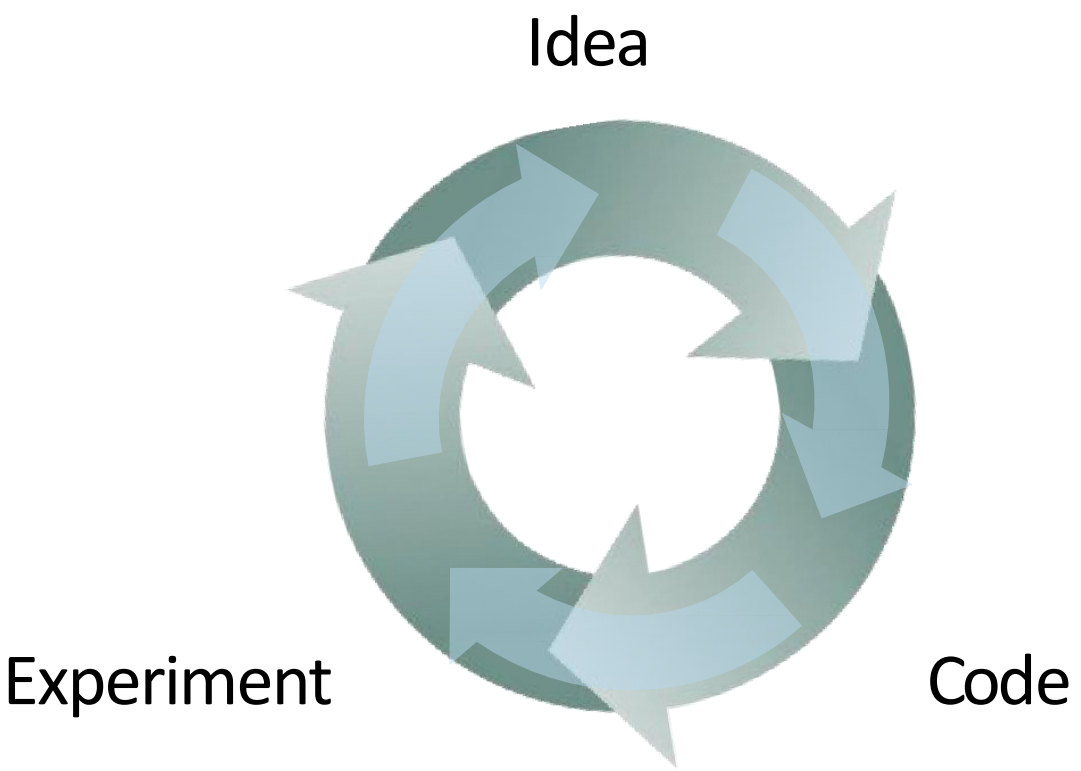


Deep Learning

This game is called *while True: learn()*

Alchemy

Applied DL is a very empirical process



Underfitting and overfitting

Underfitting and overfitting

Data complexity

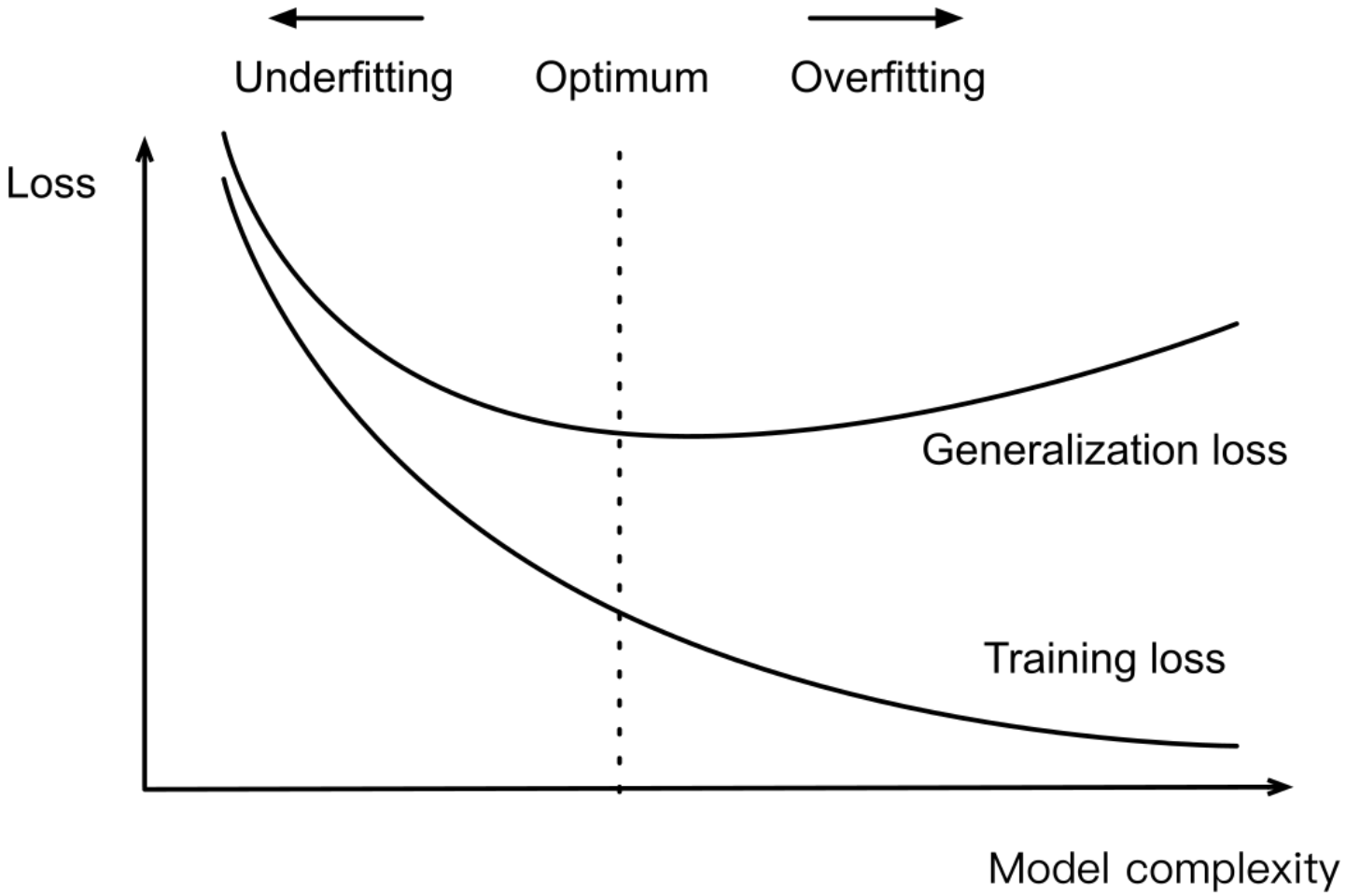
Model capacity

	Simple	Complex
Low	Normal	Underfitting
High	Overfitting	Normal

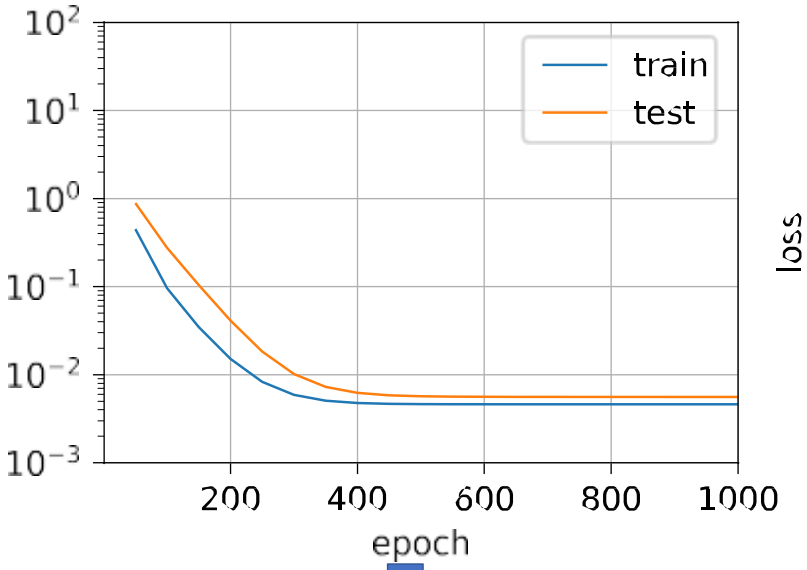
Model capacity

- The ability to fit variety of functions
- Low-capacity models struggles to fit training set
  - Underfitting
- High-capacity models can memorize the training set
  - Overfitting

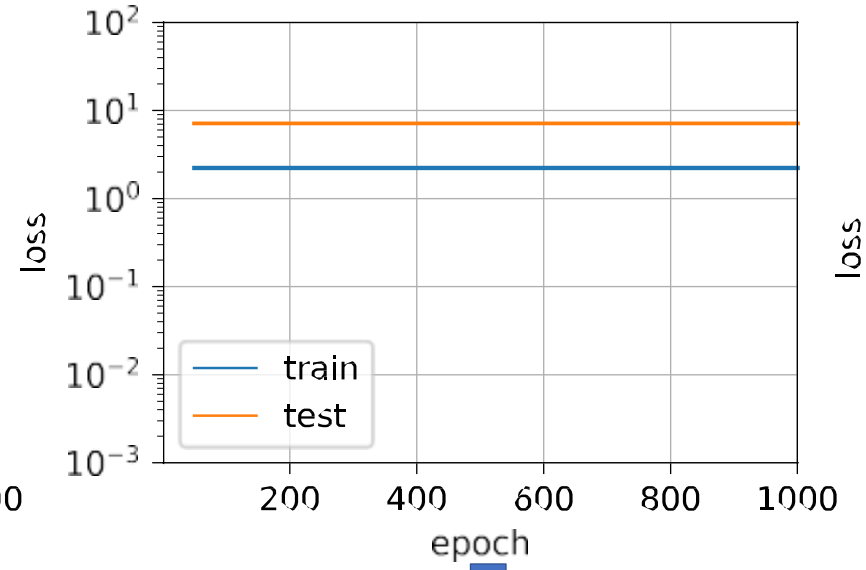
Influence of Model Complexity



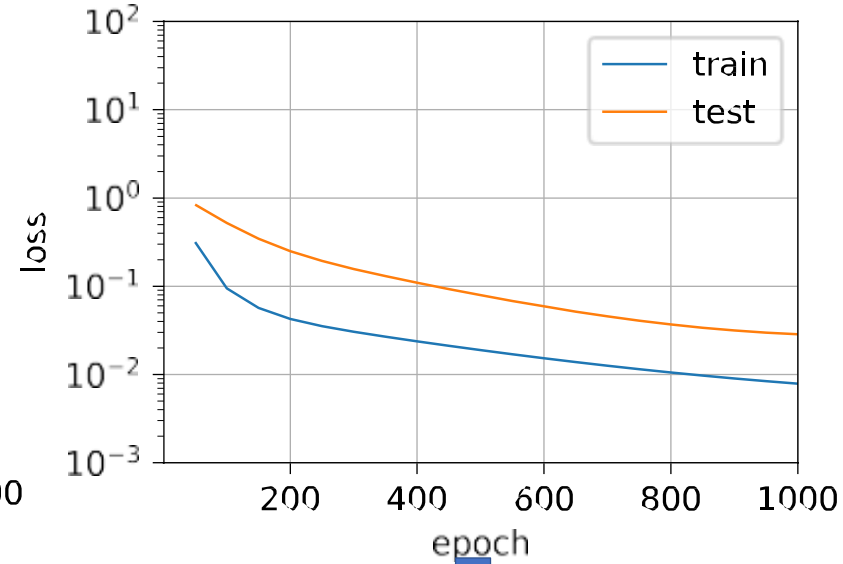
Underfitting and overfitting



desired



underfitting

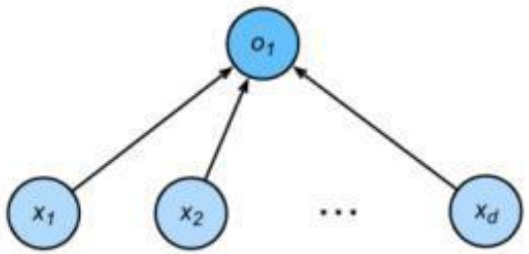


overfitting

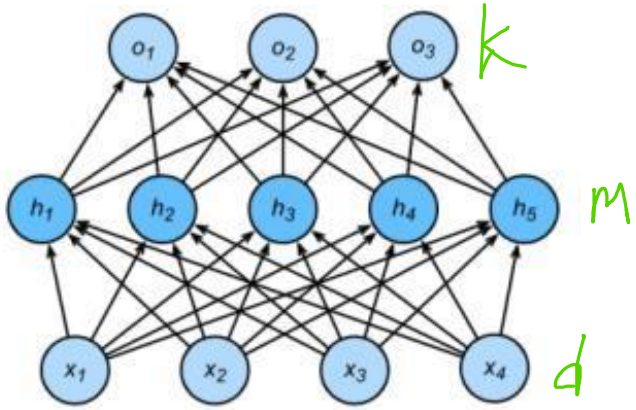
Estimate model complexity

- It's hard to compare complexity between different algorithms
  - e.g. tree vs neural network
- Given an algorithm family, two main factors matter:
  - The number of parameters
  - The values taken by each parameter

$$d + 1$$

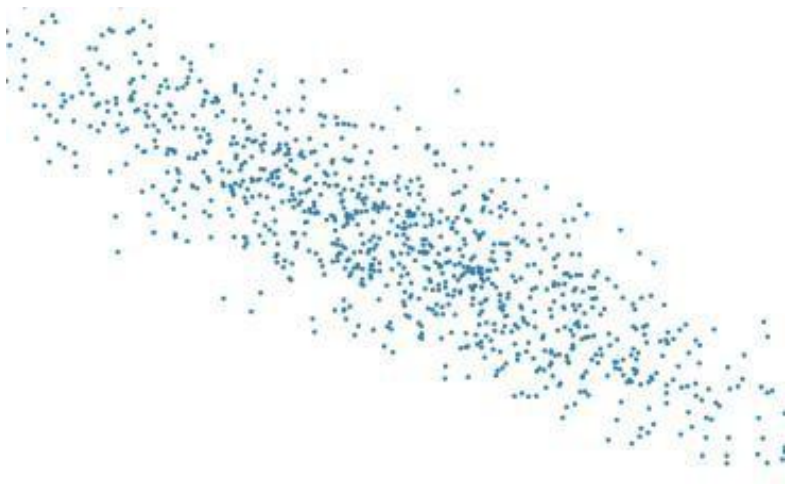


$$(d+1)m + (m+1)k$$



Data complexity

- Multiple factors matters
  - # of examples
  - # of elements in each example
  - time/space structure
  - diversity



Basic recipe for machine learning

- High bias (training data performance) → Underfitting →
  - Bigger network
  - Train longer
  - Architecture search

- High variance (validation set performance) → Overfitting →
  - More data
  - Regularization
  - Architecture search

Summary

- Model capacity and data capacity should be matched to prevent overfitting and underfitting.
- Understand fitting and bias/variance relationship
- Empirically, use training/validation error to evaluate
- Know what to do under which circumstance